Shri Vithal Education & Research Institute's
College of Engineering, Pandharpur
(An Autonomous Institute)

Drive Twin: A Personalised Driver Behaviour Twin for Real-Time Risk Assessment using Multimodal Sensor Fusion

Group No.–16

Name of Students- 1.Jay Yuvraj Gadekar

2.Chaitanya Narhari Gavali

3.Aman Sadik Shaikh

4.Rushikesh Rajkumar Jadhvar

Name of Guide- Mr. M. M. Landge

LITERATURE REVIEW

Sr. No.	Paper Title	Author Name	Publication Year	Methodology	Gap
1.	Driver Monitoring Systems: State of the Art and Future Trends	S. Martin, A. Eskandarian, et al.	2023	Surveyed camera-based Driver monitoring systems using eye tracking, head pose, estimation and facial landmark detection.	Focuses on event-based monitoring ony; lacks personalized driver behaviour modelling and long-term learning.
2.	Multi-Modal Driver State Monitoring Using Sensor Fusion	J. Wang, H. Sun, Y. Li, et al	2024	Combined camera, IMU, and vehicle telemetry to estimate driver state using multimodal sensor fusion techniques.	Detects driver state but does not create a Digital Twin or continuously adapt to individual driving behaviour.
3.	Machine Learning Based Driver Behaviour Analysis for Intelligent Transportation	Z. Li, C. Chen, X. Zhao, et al.	2024	Developed frameworks for vehicles and transportation infrastructure to improve simulation	Primarily focuses on vehicle-centric Digital Twins rather than modelling human driver behaviour in real time.

OBJECTIVE OF PROJECT

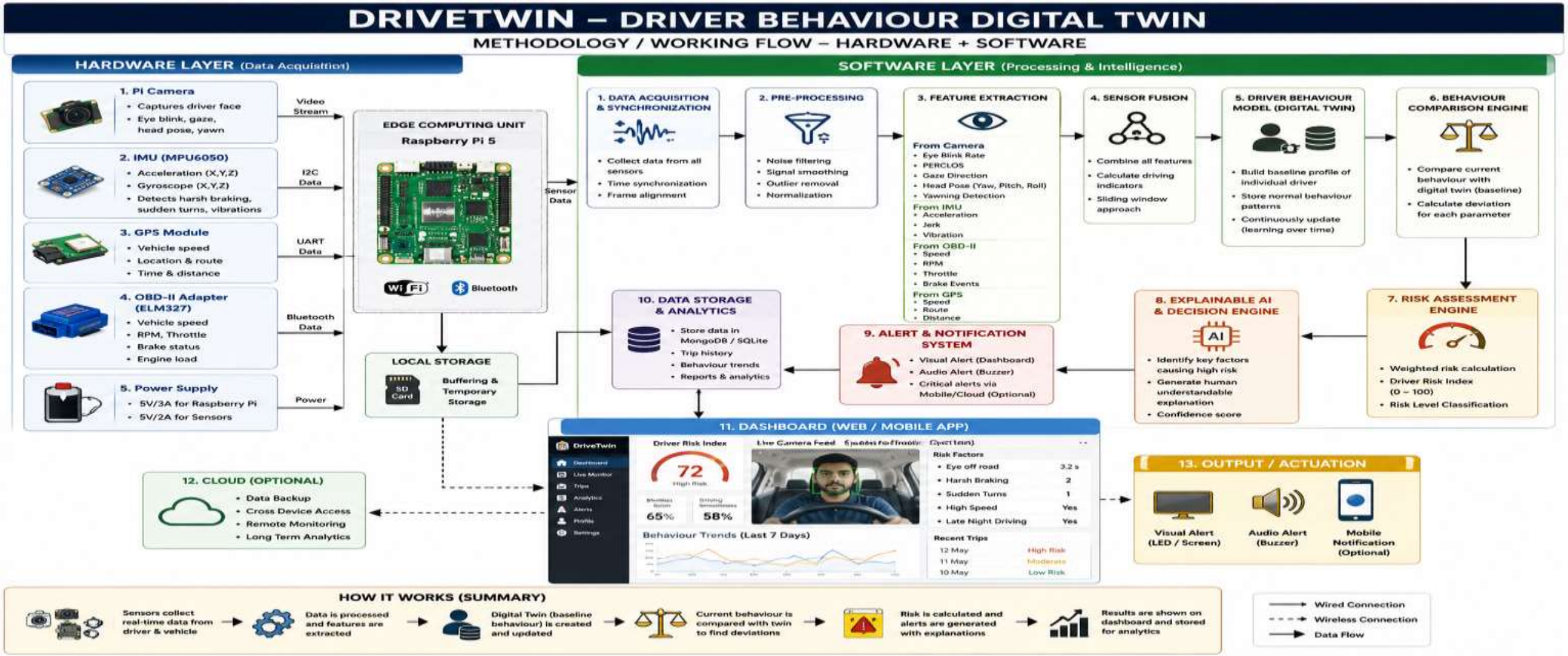
1. Develop a personalized **Driver Digital Twin** that learns individual driving behaviour.
2. Collect and fuse data from multiple sensors including camera, IMU, GPS, and vehicle parameters.
3. Analyze driving characteristics such as attention, steering behaviour, acceleration, and braking patterns.
4. Detect deviations from normal driving behaviour in real time.
5. Generate a personalized Driver Risk Index.
6. Provide explainable alerts and visual analytics through an interactive dashboard.
7. Improve road safety by enabling predictive driver assistance.

PROBLEM STATEMENT

Current Driver Monitoring Systems primarily focus on detecting isolated events such as drowsiness or distraction using fixed thresholds. These systems fail to understand individual driving behaviour, resulting in false alarms and limited adaptability. There is a need for a personalized, real-time driver behaviour modelling system that continuously learns driving patterns, detects behavioural deviations, and provides intelligent risk assessment using multimodal sensor fusion.

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METHODOLOGY



FEASIBILITY AND VIABILITY

- Uses **low-cost, readily available hardware** such as Raspberry Pi, ESP32, Pi Camera, IMU, and GPS modules.
- Developed using **open-source software** including Python, OpenCV, MediaPipe, and TensorFlow Lite.
- Performs **real-time driver behaviour analysis** through multimodal sensor fusion and edge computing.
- **Modular architecture** allows easy integration, maintenance, and future upgrades.
- Suitable for applications in **private vehicles, fleet management, driver training, and insurance telematics**.
- Provides a **cost-effective, scalable, and commercially viable** solution for next-generation personalized ADAS and intelligent transportation systems.

IMPACT

- Improves road safety through personalized driver behavior monitoring.
- Reduces false warnings by adapting to individual driving habits.
- Enables early detection of fatigue, distraction, and abnormal driving patterns.
- Assists fleet operators in evaluating driver performance and safety.
- Supports insurance companies in usage-based risk assessment.
- Provides valuable behavioral analytics for driver training and improvement.
- Serves as a foundation for future AI-enabled and human-centric ADAS technologies.

RESEARCH AND REFERENCES

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